

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026

Biology: Class 9 (Science Group)

Syllabus: Chapter 1 (The Science of Biology) | Chapter 2 (Biodiversity) | Chapter 3 (The Cell) | Chapter 4 (Cell Cycle) | Chapter 5 (Tissues, Organs and Organ Systems) | Chapter 6 (Molecular Biology) | Chapter 7 (Metabolism, complete, with the main focus on 7.7 Photosynthesis and 7.8 Respiration)

Total Marks: 38

Time: 1 Hour 45 Minutes

Paper: Exam-17 (Version 1)

Instructions:

1. Attempt ALL questions. There is no choice or OR alternative anywhere in this paper.
2. Section A (10 marks): write the question number and the letter of your answer, for example **1. C**. Check that each number matches its letter before you move on.
3. Section B (12 marks): three short questions of 4 marks. Section C (16 marks): two long questions of 8 marks.
4. Answer every part (a), (b), (c)... separately and label it. Where a question asks "why", give the reason, not just the name.
5. Write equations in full with the conditions (for example light and chlorophyll) where the textbook shows them.
6. Use black or blue ink. Pencil is allowed only for the diagram in Question 3(i).

SECTION A: Multiple Choice Questions (10 × 1 = 10 Marks)

#	Q. Choose the best possible option.	A	B	C	D	Answer
1	During the light dependent reactions, a water molecule splits (photolysis of water). According to the textbook, this splitting takes place to make up for the electron loss of the:	primary electron acceptor	NADP+	chlorophyll of photosystem II	chlorophyll of photosystem I	A B C D
2	The second phase of photosynthesis is called the light independent reactions because these reactions:	do not use light directly	take place only in the dark at night	take place in the thylakoid membranes	release oxygen out of the leaf	A B C D
3	The oxygen produced in the light dependent reactions leaves the leaf through the:	cuticle	xylem	root hairs	stomata	A B C D
4	A child left a carton on the lawn for two days. When the carton was picked up, the grass under it had turned yellow. What caused the grass to change colour?	lack of oxygen	lack of carbon dioxide	lack of light	lack of water	A B C D
5	In aerobic respiration, when 2 pyruvic acid molecules combine with 2 coenzyme A to form 2 acetyl CoA, the other products are:	2 ATP + 2 NADH	2 CO ₂ + 2 NADH	4 CO ₂ + 2 FADH ₂	6 CO ₂ + 6 H ₂ O	A B C D
6	In the electron transport chain, the electrons carried by each FADH ₂ provide enough energy to make:	two ATP molecules	three ATP molecules	four ATP molecules	thirty-six ATP molecules	A B C D
7	The earliest living organisms produced energy by anaerobic respiration because the early earth:	had no liquid water	had no free oxygen in its atmosphere	had no carbon dioxide	received no sunlight	A B C D
8	Fermentation by yeast is used in:	making yogurt from milk	making cheese	the working of skeletal muscles	wine making and baking	A B C D
9	According to the textbook, the organ of the human body that uses the maximum energy in the form of ATP (about 25% of the ATP) is the:	brain	heart	skeletal muscles	stomach	A B C D
10	At the end of the electron transport chain, the final electron acceptor (oxygen) combines with hydrogen ions to produce:	carbon dioxide	water	lactic acid	glucose	A B C D

SECTION B: Short Questions (3 × 4 = 12 Marks)

Q	Question	Marks
2(i)	A student sets up three test tubes, each with the same amount of the same pond weed in water, and counts the gas bubbles for 10 minutes. Tube P is kept in bright light and Tube Q in dim light, both at the same warm temperature. (a) Which tube, P or Q, will give more bubbles? Name the limiting factor that explains it. (1) (b) The student keeps adding sodium bicarbonate (which gives CO₂) to Tube P. The bubbles increase at first, then stop increasing. Explain why, using what the textbook says about carbon dioxide. (1) (c) At what temperature is the rate of photosynthesis highest, and what is this temperature called? (1) (d) Tube R is set up like Tube P but kept at a temperature well above this value. What happens to its rate of photosynthesis compared with P? (1)	4
2(ii)	A student runs a fast 400 metre race. Near the end, the leg muscles feel very tired. (a) Why do the leg muscles not get enough oxygen during this race? (1) (b) Which type of respiration do the muscle cells start to carry out? (1) (c) Write the equation (in words) showing what happens to pyruvic acid in these muscle cells. (1) (d) According to Fig. 7.12, how many ATP are shown from glycolysis, and how many from the electron transport chain? (1)	4
2(iii)	Answer these about the Calvin cycle (light independent reactions): (a) Carbon dioxide combines with an already existing molecule. Name it and give its number of carbon atoms. (1) (b) The intermediate formed is unstable. How many carbon atoms does it have, and what does it split into? (1) (c) Which two molecules, made in the light dependent reactions, are used to change 3PGA into G3P? (1) (d) Give the two uses of the G3P molecules. (1)	4

SECTION C: Long Questions (2 × 8 = 16 Marks)

Q	Question	Marks
3(i)	Textbook Activity 7.3 demonstrates the release of oxygen during photosynthesis using Hydrilla. (a) Draw a neat, well-labelled diagram of the set-up. Label: beaker, water with sodium bicarbonate, Hydrilla, inverted funnel, test tube, and the gas collected. (3) (b) Why is sodium bicarbonate (NaHCO₃) dissolved in the water? (1) (c) Why is the funnel slightly raised above the bottom of the beaker? (1) (d) Why is the set-up placed in sunlight for 2 to 3 hours? (1) (e) How is the collected gas tested, and what does the result show? (1) (f) In which step of the light dependent reactions is this oxygen produced? (1)	8
3(ii)	Photosynthesis and aerobic respiration are interlinked with each other. (a) Write the textbook equation of photosynthesis and the equation of aerobic respiration. (2) (b) Using your two equations, show two ways in which the products of one process are used by the other. (2) (c) Give four differences between photosynthesis and aerobic respiration in a table (for example, where each process takes place). (4)	8

Invigilator's Signature: _____

Candidate's Signature: _____

End of Question Paper